Q1. Explain the concept of an Intelligent Agent. Describe the PEAS (Performance, Environment, Actuators, Sensors) framework with one detailed example (e.g., self-driving car, medical diagnosis system, etc.).

Q2. What are Learning Agents? Explain the four main components of a learning agent with a diagram.

Q3. Discuss advantages and limitations of AI. Is AI always reliable? Justify your answer with reasoning.

Q4. Construct truth tables for the following:

a) $(p \rightarrow q) \wedge (q \rightarrow r)$

b) $(p \wedge q) \rightarrow (p \vee q)$

c) $(p \leftrightarrow q) \rightarrow (\neg p \vee q)$

Clearly show all intermediate columns.

Q5. Using laws of logic, simplify:

a) $\neg(p \vee q) \wedge (p \rightarrow q)$

b) $(p \rightarrow q) \wedge (p \rightarrow r)$

c) $\neg(p \rightarrow q)$

Show each step and mention the law used.

Q6. Translate into propositional logic:

1. If it is raining, then the ground is wet.
2. It is not raining but the ground is wet.
3. If the ground is wet, then either it rained or the sprinkler was on.

Define symbols clearly.

Q7. Prove using logical equivalence:

$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

Q8. Translate the following into Predicate Logic:

1. All students are intelligent.
2. Some students are hardworking.
3. Every intelligent student passes the exam.
4. There exists a student who does not pass.

Q9. Negate the following:

a) $\forall x$ (Student(x) $\rightarrow$ Passed(x))
b) $\exists x$ (Human(x) $\wedge$ Honest(x))
c) $\forall x \exists y$ Loves(x, y)
Show step-by-step negation.

Q10. Explain the difference between Propositional Logic and Predicate Logic with examples. Why is Predicate Logic more expressive?

Q11. Given:

1. $P \rightarrow Q$
2. $Q \rightarrow R$
3. $R \rightarrow S$
4. P

Prove: S
Use resolution method only.
Show:

- CNF conversion
- Negation of goal
- All resolvents
- Final empty clause

Q12. Given:

1. $P \vee Q$
2. $\neg P \vee R$
3. $\neg Q$

Prove: R
Apply resolution and show each resolvent.

Q13. Knowledge Base:

1. The system is secure or the firewall is active.
2. If the firewall is active, then network traffic is monitored.
3. The system is not secure.

Prove: Network traffic is monitored.
Use propositional resolution.

Q14. Convert the following into CNF:

a) $(P \rightarrow Q) \wedge (R \rightarrow S)$
b) $\neg(P \wedge (Q \rightarrow R))$
c) $(P \leftrightarrow Q)$
Show each step clearly.

Q15. Find the Most General Unifier (MGU) for:

a) P(x, y) and P(a, b)

b) Knows(John, x) and Knows(y, Mary)

c) Loves(x, f(y)) and Loves(a, f(b))

d) P(x, x) and P(a, b)

State whether unification succeeds or fails.

Q16. Given:

1. ∀x (Human(x) → Mortal(x))
2. Human(Socrates)

Prove: Mortal(Socrates)

Use resolution method.

Show:

- Implication removal
- CNF
- Negation of goal
- Unification
- Resolvents

Q17. Given:

1. ∀x (Cat(x) → Animal(x))
2. ∀x (Animal(x) → LivingBeing(x))
3. Cat(Tom)

Prove: LivingBeing(Tom)

Use resolution only.

Q18. Given:

1. ∀x (Student(x) → ∃y Likes(x, y))
2. Student(Alice)

Prove: ∃y Likes(Alice, y)

Convert to CNF and apply resolution.

Q19. What are the criteria of success for an AI system?

Discuss:

- Rationality
- Optimality
- Performance measure
- Environment assumptions

Give one example where an AI system may be intelligent but not optimal.

Q20. Differentiate between:

- Simple Reflex Agent
- Model-Based Agent

- Goal-Based Agent
- Utility-Based Agent

Give one real-world example for each.

Q21. Determine whether the following are:

- Tautology
- Contradiction
- Contingency

a) $(p \rightarrow q) \vee (q \rightarrow p)$

b) $p \wedge \neg p$

c) $(p \vee q) \rightarrow p$

Use truth table method.

Q22. Using logical equivalence laws, prove:

$$p \vee (p \wedge q) \equiv p$$

Mention the name of each law used.

Q23. Using truth table, prove:

$$p \oplus q \equiv \neg(p \leftrightarrow q)$$

Q24. Translate into Predicate Logic:

1. Every programmer knows at least one programming language.
2. Some students do not like mathematics.
3. There is a teacher whom every student respects.

Q25. Given:

1. $A \rightarrow B$
2. $B \rightarrow C$
3. $\neg C$

Prove: $\neg A$

Draw the resolution tree clearly.

Submission Deadline: March 06, 2026

Note: The handwritten assignment must be scanned and uploaded as a single PDF file. Identical submissions may be rejected and awarded zero marks.